

- Steps
9. $Ra(c) \Rightarrow \neg \neg An(c)$
 10. $\neg An(c) \Rightarrow \neg Ra(c)$
 11. $Ra(c) \Rightarrow \neg Ra(c)$
 12. $\boxed{\forall x (Ra(x) \Rightarrow \neg Ra(x))}$

- Reason
- Universal instantiation on 3
 Contrapositive on 5
 Hypothetical syllogism on 9, 10
 Universal generalization on 11 ^{Rabbits}
 2) ~~Rabbits~~ are not Rodents.

Valid conclusion:

1) All mice gnaws their food

10) For each of these sets of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.

a) "If I play hockey, then I am sore the next day." "I use the whirlpool if I am sore." "I did not use the whirlpool."

Soln: p : I play hockey q : I am sore the next day r : I use the whirlpool

Premises: 1) $p \rightarrow q$ 2) $\neg r$ 3) $\neg r$

Reason

- Steps
1. $p \rightarrow q$
 2. $q \rightarrow r$
 3. $\neg r$
 4. $\neg q$
 5. $\neg p$

Premise

Premise

Premise

Modus tollens on 3, 2

Modus tollens on 4, 1

Valid conclusion: 1) I do not play hockey 2) I am not sore.

b) "If I work, it is either sunny or partly sunny." "I worked last Monday or I worked last Friday." "It was not sunny on Tuesday." "It was not partly sunny on Friday."

Soln:

$W(x)$: I worked on day x

$S(x)$: It is sunny on day x

Domain: Set of all weekdays

$PS(x)$: It is partly sunny on day x